

Vishal Tiwari

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

PhD and MS in Physics

Aug. 2020 – Expected Jul. 2026

Thesis: Radiation Magnetohydrodynamic Simulations of Accretion Flows around Massive Black Hole Binaries

Advisor: Prof. Tamara Bogdanović

University of Massachusetts-Dartmouth, Dartmouth, MA

MS in Physics

Sep. 2018 – Jul. 2020

Thesis: Probing Type Ia Supernovae through Simulations and Observations

Advisor: Prof. Robert Fisher

International Institute of Information Technology - Hyderabad, India

B.Tech (Honours) and MS in Computer Science and Engineering

Aug. 2010 – Jul. 2017

Thesis: Geo-Visualization in 4D environment - Simulation of floods over an Urban Area

Advisor: Prof. K. S. Rajan

WORK EXPERIENCE

Graduate Research Assistant, Georgia Institute of Technology

Jan. 2022 – current.

Worked on simulations of circumbinary disk and dynamical friction

Thesis advisor: Tamara Bogdanović

Graduate Teaching Assistant, Georgia Institute of Technology

Aug. 2020 – Dec. 2021

Responsible for taking laboratory and problem sessions, holding office hours, and grading for PHYS 2211 (Intro Physics)

Graduate Research Assistant, UMass Dartmouth

Sept. 2018 – Jul. 2020

Worked on the progenitor problem of Type Ia Supernovae

Thesis advisor: Robert Fisher

Teaching Assistant, IIIT-Hyderabad

Responsible for taking tutorial sessions, managing assignment portal and grading exams for the following Computer Science courses:

- Principle of Programming Languages
- Spatial Informatics

Aug. 2013 – Dec. 2013

Aug. 2014 – Dec. 2014

Technology Associate, Morgan Stanley

Aug. 2015 – Oct. 2016

Worked with the Global Banking Team as a Java developer working on lending based services.

Software Development Intern, HackerRank

May. 2014 – July. 2014

Worked on expanding the width and depth of HackerRank Brahma Api.

Software Development Intern, Google Summer of Code

May. 2013 – Aug. 2013

Worked for Open Source Geospatial Foundation (OSGeo) for porting voronoi tessellation from JTS Topology Suite to GEOS.

GRANTS

1. PI - **Awarded 200,000 credits on NSF Advanced Cyberinfrastructure Coordination Ecosystem: Services Support (ACCESS)** - Three-Dimensional Radiative Magnetohydrodynamics Simulations of Super-Eddington Circumbinary Disks Around Massive Black Hole Binaries - **Tiwari, V.**, Chan, C. and Bogdanović, T. - April 19, 2024 - April 18, 2025.

2. co-PI - **Awarded 750,000 credits on NSF Advanced Cyberinfrastructure Coordination Ecosystem: Services Support (ACCESS)** - *Self-consistent predictions of electromagnetic signatures from minidisks in massive black hole binaries* - Chan, C., **Tiwari, V.** and Bogdanović, T. - April 19, 2024 - April 18, 2025.
3. co-PI - **Hubble Space Telescope Proposal** - *The late-time NIR light curve of SN 2018gv. HST Proposal, p.15693.* - Graur, O., Eisenstein, D.J., Fisher, R. and **Tiwari, V.** - 2019.

PUBLICATIONS

1. **Tiwari, V.**, Chan, C., Bogdanović, T., Jiang, Y., Davis, S., Radiation Magnetohydrodynamic Simulation of sub-Eddington Circumbinary Disk in a 10:1 Massive Black Hole Binary. arXiv e-prints, page arXiv:2510.13955, October 2025. (The Astrophysical Journal, submitted)
2. **Tiwari, V.**, Chan, C. H., Bogdanović, T., Jiang, Y. F., Davis, S. W., Ferrel, S. (2025). Radiation Magnetohydrodynamic Simulation of Sub-Eddington Circumbinary Disk around an Equal-mass Massive Black Hole Binary. The Astrophysical Journal, 986(2), 158.
3. Chan, C. H., **Tiwari, V.**, Bogdanović, T., Jiang, Y. F., Davis, S. W. (2025). Radiative magnetohydrodynamics simulation of minidisks in equal-mass massive black hole binaries. The Astrophysical Journal, 991(1), 71.
4. **Tiwari, V.**, Graur, O., Fisher, R., Seitzenzahl, I., Leung, S.C., Nomoto, K.I., Perets, H.B. and Shen, K., 2022. The late-time light curves of Type Ia supernovae: confronting models with observations. Monthly Notices of the Royal Astronomical Society, 515(3), pp.3703-3715.
5. Roy, N.C., **Tiwari, V.**, Bobrick, A., Kosakowski, D., Fisher, R., Perets, H.B., Kashyap, R., Lorén-Aguilar, P. and García-Berro, E., 2022. 3D Hydrodynamical Simulations of Helium-ignited Double-degenerate White Dwarf Mergers. The Astrophysical Journal Letters, 932(2), p.L24.

PAPERS in PREPARATION

1. **Tiwari, V.**, Jiang, Y., Chan, C., Bogdanović, T., Davis, S., Radiative Magnetohydrodynamics Simulations of super-Eddington Mini-Disks in 10:1 Massive Black Hole Binaries. (in prep.)
2. Chan, C. H., **Tiwari, V.**, Bogdanović, T., Jiang, Y. F., Davis, S. W., Radiative magnetohydrodynamics simulation of minidisks in unequal-mass massive black hole binaries (in prep.)
3. **Tiwari, V.**, Chan, C., Bogdanović, T., Jiang, Y., Davis, S., Magnetohydrodynamics Simulations of Circumbinary Disks around Eccentric Massive Black Hole Binaries. (in prep.)
4. Mehta, V., **Tiwari, V.**, Pakmor R., Singh, D., Fisher R., Hydrodynamical simulations of helium-ignited binary white dwarf mergers. (in prep.)

CONFERENCE PROCEEDINGS/ABSTRACTS

1. **Tiwari, V.**, Chan, C. H., Bogdanovic, T., Jiang, Y. F., Davis, S. (2024, February). First Radiation Magnetohydrodynamic Simulations of Accretion Flows around Massive Black Hole Binaries Heading for Merger. In American Astronomical Society Meeting Abstracts (Vol. 56, No. 2, pp. 229-03).
2. Chan, C. H., **Tiwari, V.**, Bogdanović, T., Jiang, Y., Davis, S. (2024, February). Emission properties of massive black hole binaries heading for merger. In American Astronomical Society Meeting Abstracts (Vol. 56, No. 2, pp. 229-04).
3. Kosakowski, D., Roy, N., **Tiwari, V.**, Bobrick, A., Fisher, R., Perets, H., Kashyap, R. (2022, June). 3D Hydrodynamical Simulations of the Dynamically Driven Double-degenerate Double-detonation Scenario of Type Ia Supernovae. In American Astronomical Society Meeting 240 (Vol. 54, No. 6, pp. 317-04).
4. Fisher, R., **Tiwari, V.**, Graur, O. (2020, June). Constraining the Progenitors and Mechanisms of Type Ia Supernovae Through Late-Time Light Curve Observations. In American Astronomical Society Meeting Abstracts 236 (Vol. 236, pp. 320-01).
5. **Tiwari, V.**, Fisher, R. T., Graur, O. (2018). Constraining Type Ia Supernovae with Models and Observations of Late-Time Light Curves. Bulletin of the American Physical Society, 63.

TALKS and POSTERS

1. **NANOGrav's Astrophysics Working Group, Remote, May 30th, 2025** - Talk on *"Radiation Magnetohydrodynamic Simulations of sub-Eddington Circumbinary and Mini Disks around an Equal-mass Massive Black Hole Binary"*
2. **Unveiling massive black hole evolution with gravitational waves and light, Paris, France, May 19th-23rd, 2025, Remote** - Talk on *"Insights into Emission from Circumbinary Disks around Merging Massive Black Hole Binaries via Radiation Magnetohydrodynamic Simulations"*
3. **The Era of Binary Supermassive Black Holes: Coordination of Nanohertz-Frequency Gravitational-Wave Follow-up, Aspen, CO, February 2nd-7th, 2025** - Poster on *"Radiation Magnetohydrodynamic Simulations of Circumbinary Disks around Merging Massive Black Hole Binaries"*
4. **The Formation and Early Evolution of Supermassive Black Holes, Baltimore, MD, November 20-22nd, 2024** - Poster on *"Radiation Magnetohydrodynamic Simulations of Circumbinary Disks around Merging Massive Black Hole Binaries"*
5. **Nevada Center for Astrophysics Symposium, Las Vegas, March 1st 2024** - Talk on *"First Radiation Magnetohydrodynamic Simulations of Accretion Flows around Massive Black Hole Binaries Heading for Merger"*
6. **Georgia Regional Astronomy Meeting, Atlanta, November 4th 2023** - Talk on *"Emission properties of massive black hole binaries heading for merger"*
7. **22nd Eastern Gravity Meeting, May 31st, 2019** - Talk on *"Dimensional Dynamically Driven Double-Degenerate Double-Detonation Simulations for Type Ia Supernova."*
8. **High Performance Computing Day, UMass Lowell, May 21, 2019** - Poster on *"Three Dimensional Dynamically Driven Double-Degenerate Double-Detonation Simulations for Type Ia Supernova."*

SERVICE

1. **Students Mentored**
 - Aaron Bartleson (Undergraduate student at Georgia Tech)
 - Divyanshu Singh (Masters student at UMass Darmouth)
 - Caleb J Derochea (Masters student at UMass Darmouth)
2. **Referee** - Astrophysics and Space Science, Springer.

SUMMER SCHOOLS

1. *Flatiron Computational Fluid Dynamics for Astrophysics Summer School*, July 31 - Aug 11, 2023, New York, NY
2. *International High-Performance Computing Summer School*, July 18 - 30, 2021, Virtual.
3. XSEDE HPC Workshop - Summer Boot Camp - June 3-6, 2019 from Boston University, MA.

AWARDS and ACHIEVEMENTS

1. First-Year Doctoral Fellowship at the University of Massachusetts, Dartmouth, 2018-2019.
2. Awarded travel allowance from IEEE Travel Award-IGARSS to give an oral presentation at the International Geoscience and Remote Sensing Symposium (IGARSS) 2015 in Milan, Italy.
3. All-India-Rank 3679 in AIEEE 2010 amid a total of over 1,000,000 students.
4. All-India-Rank 8684 in IIT-JEE 2010 of over 450,000 students.

TECHNICAL SKILLS

Astrophysical Simulation Tools: Athena++, FLASH, MESA, GIZMO

Programming Languages/Scripting: C/C++, Fortran, Python, Matlab, Java, Bash, Javascript

Data Analysis and Visualization Tools/libraries: Matplotlib, yt, VisIt, OpenGL, Processing

HPC Systems used: NASA-Pleiades, TACC-Stampede3, GaTech-PACE, UMass Dartmouth-Carnie